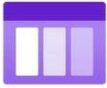


Describing Storage Accounts



Azure Queue - Message-based storage for microservices

<https://<storage-account>.queue.core.windows.net>



Azure Table - Non-relational, semi-structured data storage service

<https://<storage-account>.table.core.windows.net>



Azure Files - Cloud-based file-sharing service

<https://<storage-account>.file.core.windows.net>



Azure Blob - Object-oriented storage solution

<https://<storage-account>.blob.core.windows.net>

Components of Storage Accounts

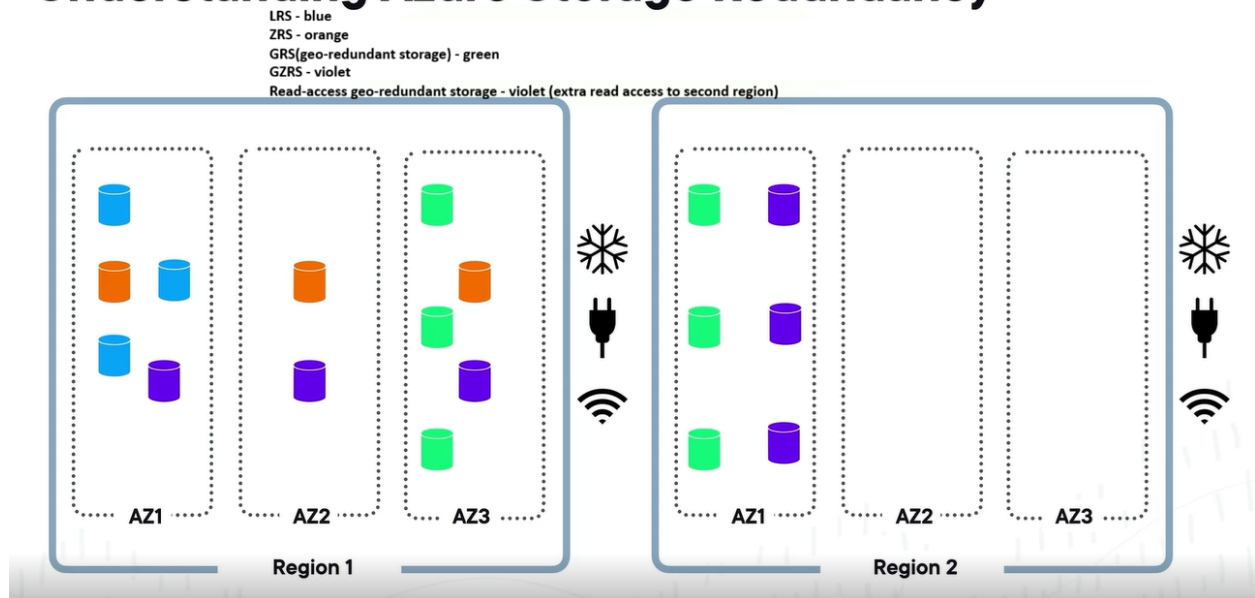
Account Type

Performance Tier

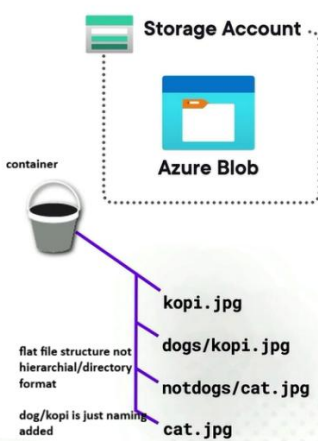
Replication

Access Tier

Understanding Azure Storage Redundancy



Describing Azure Blob Storage



Azure Blob is a sub-service or sub-resource of Azure Storage (storage accounts).

Blob storage is object based and easily accessible from HTTP(S)/REST.

Store items such as:

- Image and video files
- Text files
- Log files (append blobs)
- VHD (Virtual Hard Disk) files (page blobs)

Types of Blobs



Block Blobs
Storing images or videos.
Best suited for streaming



Append Blobs
Log Files



Page Blobs
Virtual Machine Disks

storage account name should be unique globally.

performance:

1. general purpose v2

2. premium:

type1- block blobs (best for high transaction rates or low storage latency)

type2- file shares (best for enterprise or high-performance apps that need to scale)

type3- page blobs (best for random read write operations)

access tier(for blob storage) - hot(frequent data)/cool(backup or infrequent data)

--

storage tiers (storage cost decreases and access cost increases from hot to archive)

- hot tier (default)

- cool tier

- cold tier

- archive tier

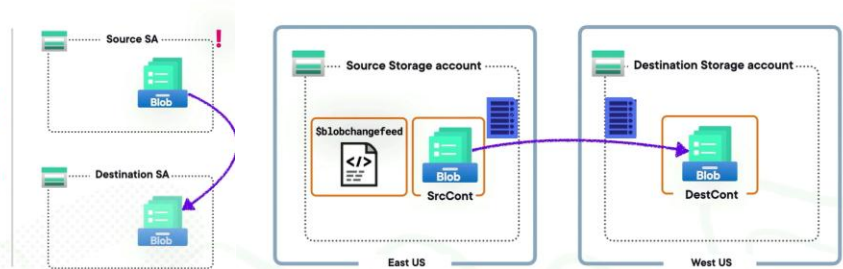
can change the tier in the container level.

blob versions (to enable storage account settings --> data management --> tracking --> enable versioning for blobs, maintain previous versions of objects and can recover if modified/deleted by mistake, when data protection enabled and versioning enabled, every modification of file creates a new version, we can promote/delete previous version to current version, when we promote a older version to current version, it will create a new version during promotion.

Describing Object Replication

Object replication asynchronously copies Block Blobs between storage accounts:

- Requires source and destination storage accounts
- Requires versioning and change feed
- Supports cross-tenant replication



Benefits of Object Replication



Minimized Latency



Increased Efficiency



Data Distribution



Cost Optimization

Tracking

☒ Enable versioning for blobs

Use versioning to automatically maintain previous versions of your blobs. [Learn more](#)

Consider your workloads, their impact on the number of versions created, and the resulting costs. Optimize costs by automatically managing the data lifecycle. [Learn more](#)

- ☒ Keep all versions
- ☐ Delete versions after (in days)

☒ Enable blob change feed

Keep track of create, modification, and delete changes to blobs in your account. [Learn more](#)

- ☒ Keep all logs
- ☐ Delete change feed logs after (in days)

Change feed not required on the destination storage account.

Create replicaton rules in the source storage account → object replication → replication rules

Microsoft Azure Search resources, services, and docs (G+)

cloud_user_p_3f1ad766... PLURALSIGHT CLOUD

Home > Storage accounts > awesomestg01 | Object replication >

Create replication rules

When you create object replication rules, blob change feed and blob versioning are automatically enabled for the source and destination storage accounts. To begin replicating objects, specify the source storage account and the destination storage account. [Learn more about copying objects in object replication](#)

Destination subscription * P5-Real Hands-On Labs

Destination storage account * awesomestg02 [Don't see your account?](#)

Container pair details

A container pair consists of a container in the source account and a container in the destination account. Objects in the source container are copied over to the destination container according to the replication rule. You can optionally filter which objects are copied by specifying a prefix match and by copying objects created only after a specified date and time.

Source container	Destination container	Filters	Copy on
srccont	destcont	0 (add)	Only new
Select a source container	Select a destination container		

Add prefix matches

A prefix match will find items like folders and blobs under the specified container that start with the specified input. For example, inputting "a" would filter any folders or blobs that start with "a". If multiple prefixes are specified, items that have any of these prefixes will be included.

Prefix match

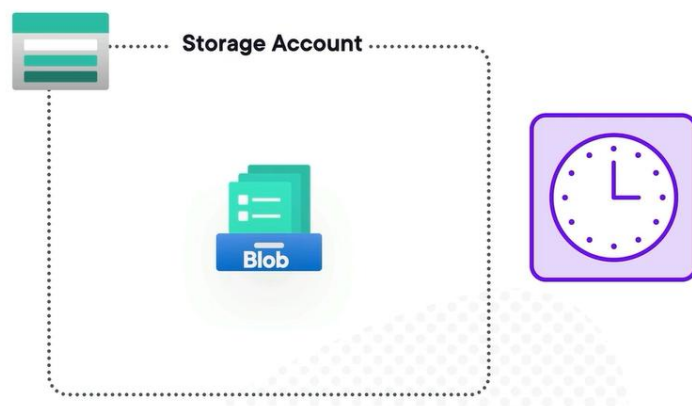
awesomestg02/

Enter a prefix

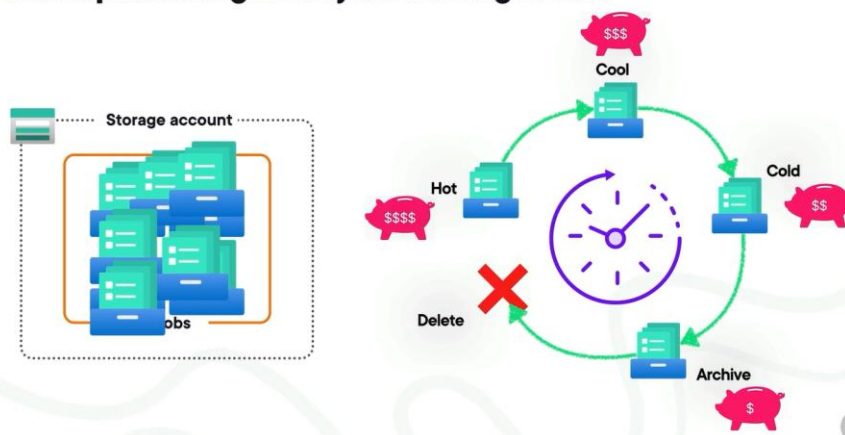
Life cycle management, as the blobs age the tier will change as per this.

Azure Blob Storage service feature that enables automation to manage lifecycle operations of Blobs.

- Automate Blob Lifecycle
- Move Access Tiers
- Optimize Costs



Conceptualizing Lifecycle Management



Add a rule ...

- 1 Details** **2** Base blobs **3** Filter set

A rule is made up of one or more conditions and actions that apply to the entire storage account. Optic rules will apply to particular blobs by limiting with filters.

Rule name *

move-to-cool

Rule scope *

- ☐ Apply rule to all blobs in your storage account
- ☒ Limit blobs with filters

Blob type *

- ☒ Block blobs
- ☒ Append blobs

Blob subtype *

- ☒ Base blobs
- ☐ Snapshots
- ☐ Versions

Add a rule ...

✓ Details 2 Base blobs 3 Filter set

Lifecycle management uses your rules to automatically move blobs to cooler tiers or to delete them. If you create multiple rules, the associated actions must be implemented in tier order (from hot to cool storage, then archive, then deletion).

If

Base blobs were *

☒ Last modified

☐ Created

More than (days ago) *

30

↓

Then

Move to cool storage

↓

+ Add conditions

Move to cool storage
For infrequently accessed data that you want to keep on cool storage for at least 30 days.

Move to cold storage
For rarely accessed data that you want to keep for at least 90 days.

Move to archive storage
Use if you don't need online access and want to keep the object for 180 days or longer.

Delete the blob
Deletes the object per the specified conditions.

Delete the blob

↓

If

Base blobs were *

☒ Last modified

☐ Created

More than (days ago) *

60

↓

Then

Move to cold storage

[Home](#) > [Storage accounts](#) > [awesomestg01](#) | [Lifecycle management](#) >

Add a rule ...

✓ Details ✓ Base blobs **3 Filter set**

Blob prefix

Filter blobs by name or first letters. To find items in a specific container, enter the name of the container followed by a forward slash, then the blob name or first letters. For example, to show all blobs starting with "a", type: "mycontainer/a".

Blob prefix



Blob index match

If you have indexed items in containers with keys and values, you can filter for them.

Key

Value





Azure Storage Encryption

By default, all data stored (data at rest) in any Azure Storage service is secured using Storage Service Encryption (SSE).

All data in transit can be secured using transport-level security (HTTPS).

To access data we use below authentication: supported for blobs, queues and tables and for files there is other options available.

Authentication Methods

Access Keys:

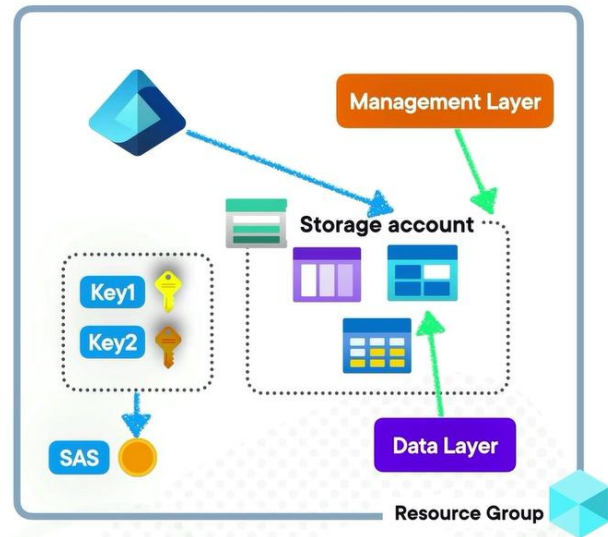
- Azure-generated keys that provide unlimited access to both the management and data layer of an Azure Storage solution.

Shared Access Signature (SAS):

- An access signature, generated from access keys, that provides limited access at either the account level or the service level.

Microsoft Entra Authentication

- Uses Azure role-based access control (RBAC) and Microsoft Entra identities to provide authentication (instead of access keys).



[Home](#) > [Storage accounts](#) >

Create a storage account

Basics Advanced Networking Data protection Encryption Tags Review + create

Encryption type * ⓘ

☒ Microsoft-managed keys (MMK)

☐ Customer-managed keys (CMK)

Enable support for customer-managed keys ⓘ

☒ Blobs and files only

☐ All service types (blobs, files, tables, and queues)

⚠ This option cannot be changed after this storage account is created.

Enable infrastructure encryption ⓘ

☐

For CMK you need to store key in keyvault and refer that here.

Create a storage account

☒ Select a key vault and key

☐ Enter key from URI

Subscription

P5-Real Hands-On Labs

Key store type

☒ Key vault

☐ Managed HSM ⓘ

Key vault

Select a key vault

[Create new](#)

Key *

Select a key

[Create new](#)

User-assigned identity * ⓘ

Select an identity

✖ Key vault details required

Enable infrastructure encryption ⓘ

☒

⚠ This option cannot be changed after this storage account is created.

Infrastructure encryption is another level of encryption on top of this.

Home / Storage accounts / awesomestg01

» **awesomestg01 | Configuration** ⚙️ ☆ ...

Storage account

«
Save
Discard
Refresh
Give feedback

- Object replication
- Blob inventory
- Static website
- Lifecycle management
- Azure AI Search

Settings

- Configuration**
- Data Lake Gen2 upgrade
- Resource sharing (CORS)
- Advisor recommendations
- Endpoints
- Locks

Monitoring

- Insights
- Alerts

The cost of your storage account depends on the usage and the options you choose below. [Learn more about storage pricing](#)

Account kind

StorageV2 (general purpose v2)

Performance ⓘ

☒ Standard ☐ Premium

ⓘ This setting cannot be changed after the storage account is created.

Secure transfer required ⓘ

☐ Disabled ☒ Enabled

Allow Blob anonymous access ⓘ

☒ Disabled ☐ Enabled

Allow storage account key access ⓘ

☐ Disabled ☒ Enabled

Allow recommended upper limit for shared access signature (SAS) expiry interval ⓘ

☒ Disabled ☐ Enabled

Default to Microsoft Entra authorization in the Azure portal ⓘ

☒ Disabled ☐ Enabled

Home > Storage accounts > awesomestg01

awesomestg01 | Shared access signature

Storage account

Search Give feedback

Data storage

- Containers
- File shares
- Queues
- Tables

Security + networking

- Networking
- Front Door and CDN
- Access keys
- Shared access signature**
- Encryption
- Microsoft Defender for Cloud

Data management

- Storage tasks (preview)
- Redundancy

A shared access signature (SAS) is a URI that grants restricted access rights to Azure Storage resources. You can provide a shared access signature to clients that should not be trusted with your storage account key but whom you wish to delegate access to certain storage account resources. By distributing a shared access signature URI to these clients, you grant them access to a resource for a specified period of time.

An account-level SAS can delegate access to multiple storage services (i.e. blob, file, queue, table). Note that stored access policies are currently not supported for an account-level SAS.

[Learn more about creating an account SAS](#)

Allowed services

☒ Blob ☒ File ☒ Queue ☒ Table

Allowed resource types

☐ Service ☐ Container ☐ Object

Allowed permissions

☒ Read ☒ Write ☒ Delete ☒ List ☒ Add ☒ Create ☒ Update ☒ Process ☒ Immutable storage ☒ Permanent delete

Blob versioning permissions

☒ Enables deletion of versions

Allowed blob index permissions

☒ Read/Write ☒ Filter

Start and expiry date/time

Start: 10:06:05 AM
End: 6:06:05 PM
(UTC-05:00) Eastern Time (US & Canada)

Allowed IP addresses

For example, 168.1.5.65 or 168.1.5.65-168.1.5.70

Allowed protocols

☒ HTTPS only ☐ HTTPS and HTTP

Preferred routing tier

☒ Basic (default) ☐ Microsoft network routing ☐ Internet routing

Some routing options are disabled because the endpoints are not published.

Signing key

key1

Generate SAS and connection string

For a particular type of data like only blob, go to that container settings or that blob in container for more specific setting in storage account.

Home > Storage accounts > awesomestg01 | Containers > awesomecontainer

awesomecontainer | Shared access tokens

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Access policy

Properties

Metadata

A shared access signature (SAS) is a URI that grants restricted access to an Azure Storage container. Use it when you want to grant access to a specific time range without sharing your storage account key. [Learn more about creating an account SAS](#)

Signing method

☒ Account key ☐ User delegation key

Signing key ⓘ

Key 1

Stored access policy

None

Permissions * ⓘ

Read

Start and expiry date/time ⓘ

Start

10:12:21 AM

(UTC-05:00) Eastern Time (US & Canada)

Expiry

6:12:21 PM

(UTC-05:00) Eastern Time (US & Canada)

Azure Files

Azure Files is a sub-service/sub-resource of Azure Storage (storage accounts).

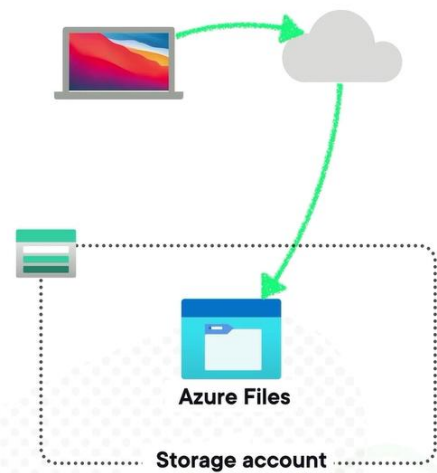
Azure Files is a managed file share service.

Features:

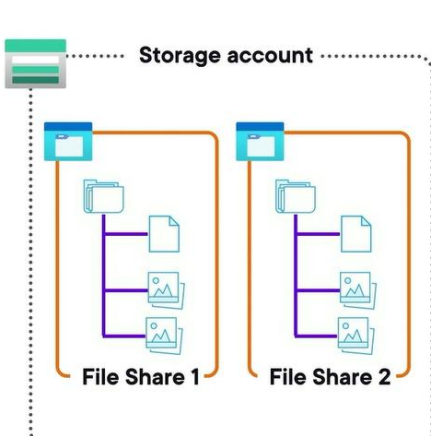
SMB/NFS connectivity

Supports Windows, Linux, and macOS

Traditional file structure



Components of Azure Files



File Service:

- Azure Files service is a sub-service of Azure Storage accounts

File Share:

- The file structure we are connecting to locally

Files and Folders:

- The files and folders that exist in the file share

Home > awesomestg01_1711637823104 | Overview > awesomestg01 | File shares >

New file share

Basics Backup Review + create

Name *

awesomeshare

Tier *

Transaction optimized

Premium

Premium file shares are only available for premium file storage accounts.

Transaction optimized

Enable transaction-heavy workloads that don't need premium-level latency. Great for applications that require file storage as a backend store.

Hot

Optimized for general purpose file sharing scenarios such as team shares and Azure File Sync.

Cool

Optimized for online archive storage scenarios.



You can improve performance and maximum share capacity by enabling large file shares for this storage account. [Learn more](#)

Azure File Snapshots

Capture system data at a specific point in time

Azure Files Snapshots

- Read-only
- Entire File Share
- Incremental



Soft Delete for Azure Files

Helps protect from accidental deletion

Entire File System

When enabled

Retained for 1-365 days

Individual files not protected

Home > Storage accounts >

Create a storage account ...

Protect your data from accidental or erroneous deletion or modification.

- ☐ Enable point-in-time restore for containers
Use point-in-time restore to restore one or more containers to an earlier state. If point-in-time restore is enabled, then versioning, change feed, and blob soft delete must also be enabled. [Learn more](#)
- ☒ Enable soft delete for blobs
Soft delete enables you to recover blobs that were previously marked for deletion, including blobs that were overwritten. [Learn more](#)
- Days to retain deleted blobs ⓘ
- ☒ Enable soft delete for containers
Soft delete enables you to recover containers that were previously marked for deletion. [Learn more](#)
- Days to retain deleted containers ⓘ
- ☒ Enable soft delete for file shares
Soft delete enables you to recover file shares that were previously marked for deletion. [Learn more](#)
- Days to retain deleted file shares ⓘ

For soft delete, Complete file share restored, not just single file. Whereas for snapshots individual files are there to restore.

Home > Storage accounts > awesomestg01 | File shares > awesomestg01 | File shares > awesomefileshare | Snapshots >

awesomefileshare (T14:24:25.000000Z) ...

File share snapshot

[Connect](#) [Refresh](#) [Properties](#)

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search files by prefix

Name	Type	Size
awesomefolder	Directory	
AwesomeFile01.txt	File	36 B
	File	28 B

Restore AwesomeFile01.txt

The snapshot file will be restored to the same directory as the original file share. A new directory will be created if the original one no longer exists.

☐ Restore as a copy and rename ☒ Overwrite original file

Restored file name

OK

Cancel

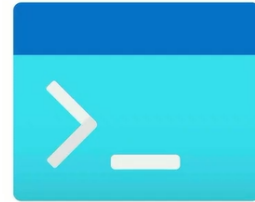
For file share level, not file level or directory level, we can use identity based authentication access, we can use windows ACLs for that.

Describing Storage Utilities



Azure Storage Explorer

A graphical user interface (GUI) tool for working with storage accounts. Support for Windows, Linux, and macOS.



AzCopy

A command-line utility for working with storage accounts. Support for Windows, Linux, and macOS.